AWS INTERNSHIP PROJECT REPORT

VPC Peering with Public and Private Subnets

Submitted by	Pitche Eshwar
Project	AWS VPC Peering
AWS Region	Europe (Stockholm) - eu-north-1
AWS Account ID	381466707358
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1. Abstract

This project demonstrates the configuration of Amazon Virtual Private Cloud (Amazon VPC) networking with two VPCs connected through VPC Peering. The environment includes subnets, EC2 instances, route tables, security group rules and connectivity testing. The AWS console screenshots in Appendix A show the VPC creation workflow, the peering connection, route-table work, security group configuration, EC2 instances and terminal-based ping tests.

2. Objectives

- Create and inspect two separate VPCs using non-overlapping IPv4 CIDR ranges.
- Configure public/private subnet resources as required by the project design.
- Create a VPC peering connection between the two VPCs.
- Configure routes on both sides so traffic can use the peering connection.
- Configure security group rules for the required test traffic.
- Launch EC2 instances and test private-network connectivity.

3. Environment and Resources

All resources were created in the Europe (Stockholm) Region, **eu-north-1**. The key resources identified in the screenshots are summarised below.

Resource	Details
Internship-VPC-1	IPv4 CIDR 10.0.0.0/16; public subnet (eu-north-1a) with internet gateway route, and a private subnet
Internship-VPC-2	IPv4 CIDR 10.1.0.0/16; private subnet (eu-north-1a) with its own route table
Peering connection	pcx-025209397f4bdb51f (Internship-VPC-1-to-VPC-2) - status Active
EC2 instance (VPC 1)	VPC1-Public-EC2 (i-0f27271142aeebe9f), private IP 10.0.1.251
EC2 instance (VPC 2)	Test target, private IP 10.1.1.196
Security groups	VPC1-Public-EC2-SG (sg-06aedfbb1f00ed626) and the VPC 2 instance security group

4. Implementation Procedure

1. **Create VPCs and subnets:** Create two VPCs with non-overlapping CIDR ranges (10.0.0.0/16 and 10.1.0.0/16) and the required public/private subnets (Figures 1, 3).
2. **Configure internet access:** Attach an internet gateway to VPC 1 and add a default route (0.0.0.0/0) to it in the public route table (Figure 9).
3. **Create VPC peering:** Request a peering connection between VPC 1 and VPC 2, accept it, and confirm that its status is Active (Figures 4, 7, 10).
4. **Configure routes:** Add a route for the peer VPC CIDR that targets the peering connection in the route tables on both sides (Figures 17, 19).
5. **Configure security groups:** Allow only the protocol and source ranges required for the test - ICMP from the peer VPC CIDR - instead of opening unnecessary inbound traffic (Figures 14, 15).

6. **Launch and test EC2 instances:** Confirm the instances are running and test communication using private IP addresses with ping (Figures 12, 13, 20).

5. Results and Verification

The screenshots document the full configuration and the troubleshooting that was needed before the test succeeded.

Stage	Observation	Figure
Peering request	First request pcx-0810c4935a9971465 showed *Pending acceptance*; adding a route to it failed with 'peering connection ID does not exist'.	4, 8
Peering established	Active connection pcx-025209397f4bdb51f (Internship-VPC-1-to-VPC-2) was created between 10.0.0.0/16 and 10.1.0.0/16.	7, 10
Routing	VPC 1 public route table has 10.1.0.0/16 -> peering connection; VPC 2 private route table has 10.0.0.0/16 -> peering connection.	17, 19
Security groups	ICMP allowed from the peer VPC CIDR on both sides (10.1.0.0/16 and 10.0.0.0/16).	14, 15
Test before fix	ping 10.1.1.196 from 10.0.1.251 gave 4 packets sent, 0 received, 100% packet loss.	12, 13
Test after fix	ping 10.1.1.196 gave 4 packets transmitted, 4 received, 0% packet loss (avg rtt about 0.36 ms).	20

The final terminal output (Figure 20) shows successful replies from the private IP address 10.1.1.196 in VPC 2 to the instance 10.0.1.251 in VPC 1, which confirms end-to-end connectivity over the peering connection.

6. Conclusion

The project demonstrates the main AWS networking steps for connecting separate VPC networks using VPC Peering. The configuration depends on an Active peering connection, correct routes on both sides, and security rules that permit the required traffic. The initial ping failed with 100% packet loss; after the peering connection was established, routes were added on both sides and ICMP was allowed in the security groups, the ping succeeded with 0% packet loss.

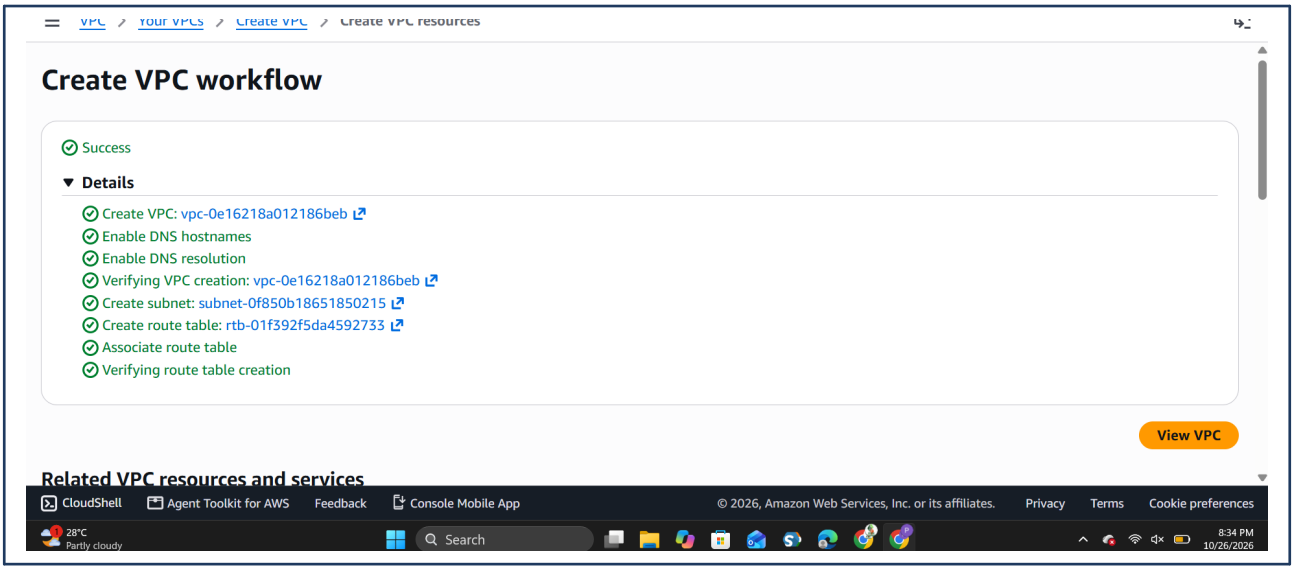
7. Final Validation Checklist

Check	Status
Peering connection pcx-025209397f4bdb51f is Active	Done (Fig. 10)
Both route tables point to this peering connection for the opposite VPC CIDR	Done (Fig. 17, 19)
EC2 instances use private IP addresses for testing	Done (Fig. 20)
Security group rules allow the exact test protocol (ICMP)	Done (Fig. 14, 15)
Terminal output clearly shows the test result	Done (Fig. 20)
Unused/accidental peering connection removed	Done (Fig. 7 lists one connection)

Appendix A - AWS Console Evidence

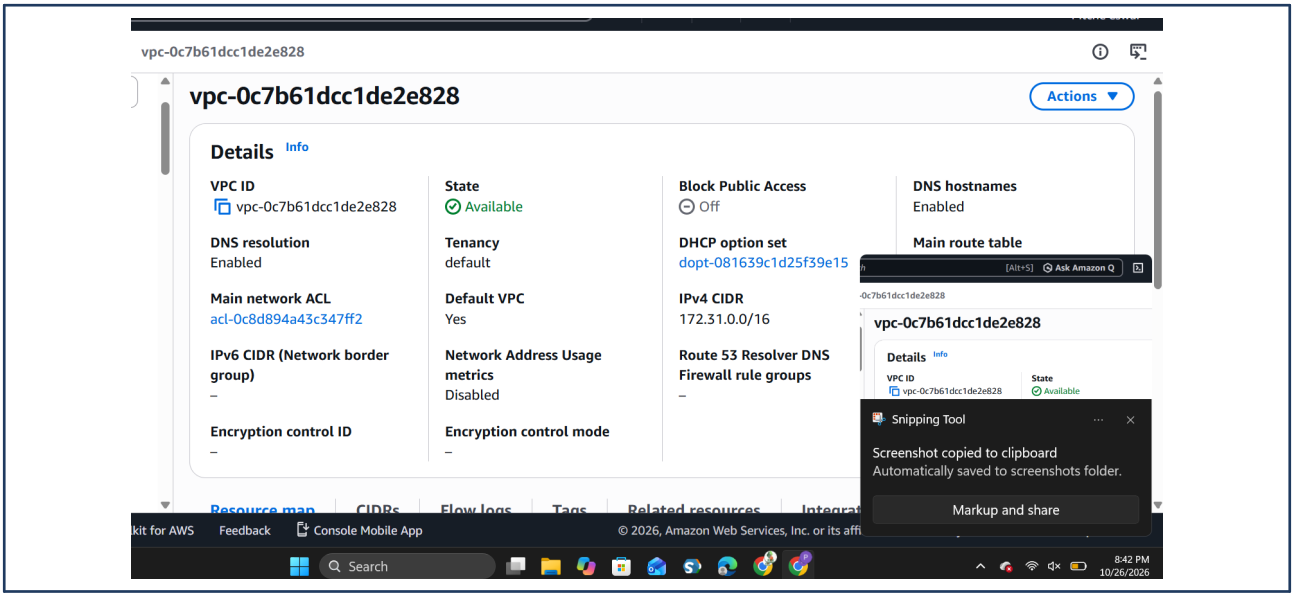
The figures below are the AWS console screenshots captured during the project. Each caption describes what is visible in the screenshot.

Figure 1. VPC creation workflow



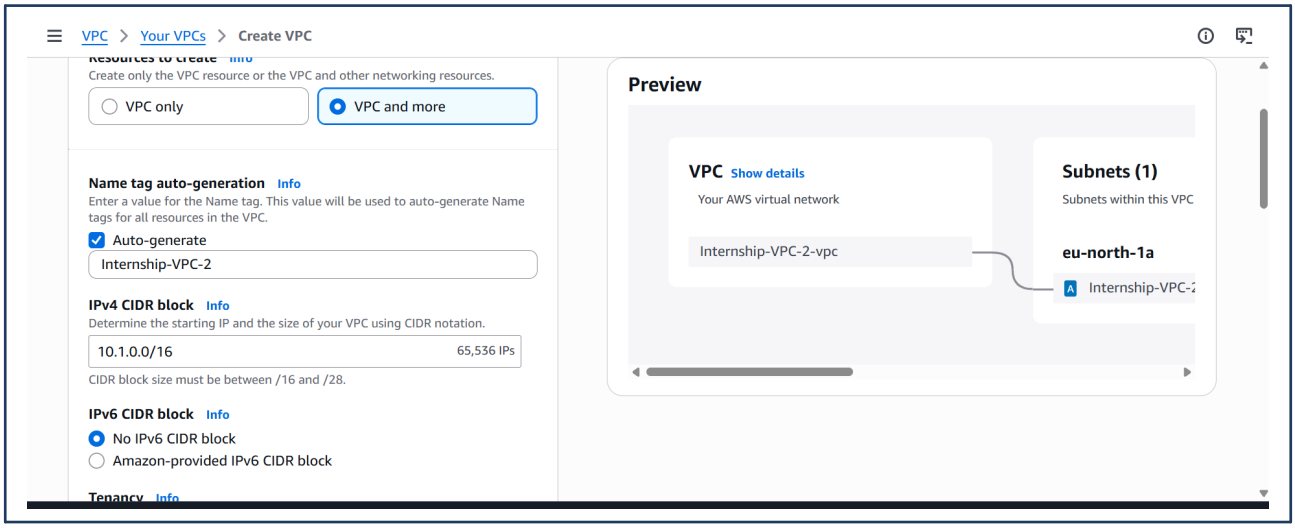
VPC creation succeeded: VPC, DNS hostnames and resolution, subnet and route table were created and associated (vpc-0e16218a012186beb).

Figure 2. VPC details (default VPC)



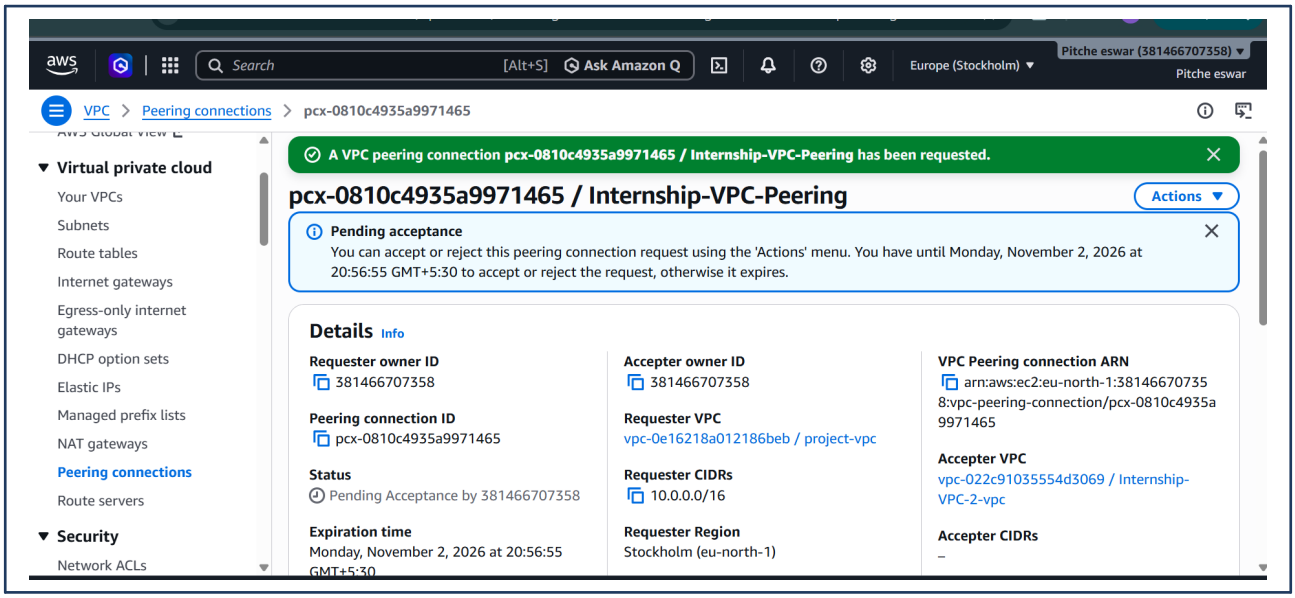
Details page of vpc-0c7b61dcc1de2e828 - State Available, IPv4 CIDR 172.31.0.0/16 (default VPC), DNS hostnames and DNS resolution enabled.

Figure 3. Creating Internship-VPC-2



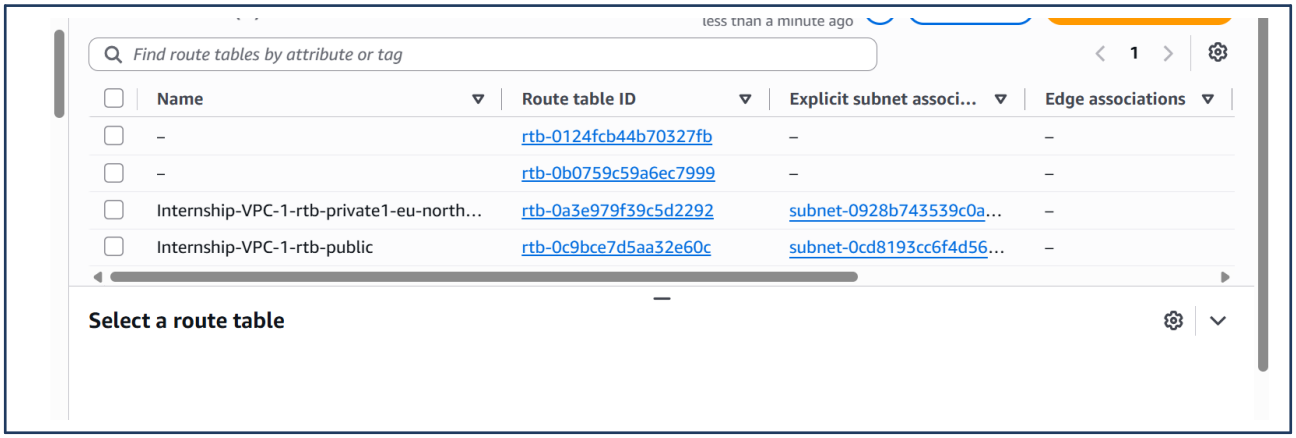
'VPC and more' wizard with name tag Internship-VPC-2, IPv4 CIDR block 10.1.0.0/16 and a preview showing one subnet in eu-north-1a.

Figure 4. Initial peering connection request



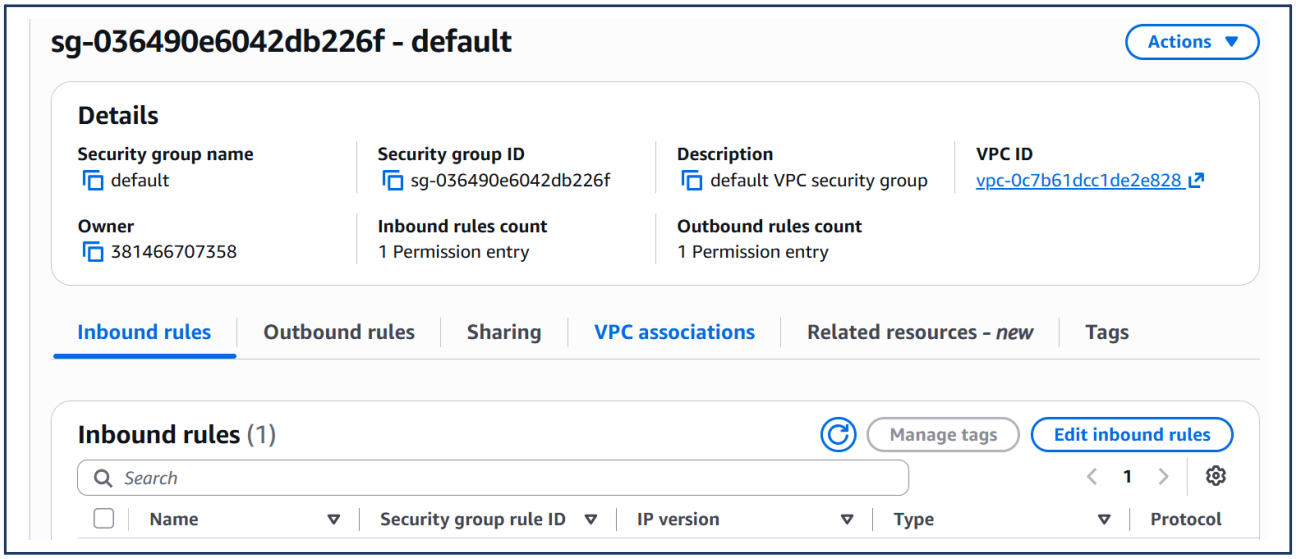
pcx-0810c4935a9971465 (Internship-VPC-Peering) between 10.0.0.0/16 and Internship-VPC-2-vpc; status shown is Pending acceptance.

Figure 5. Route tables list



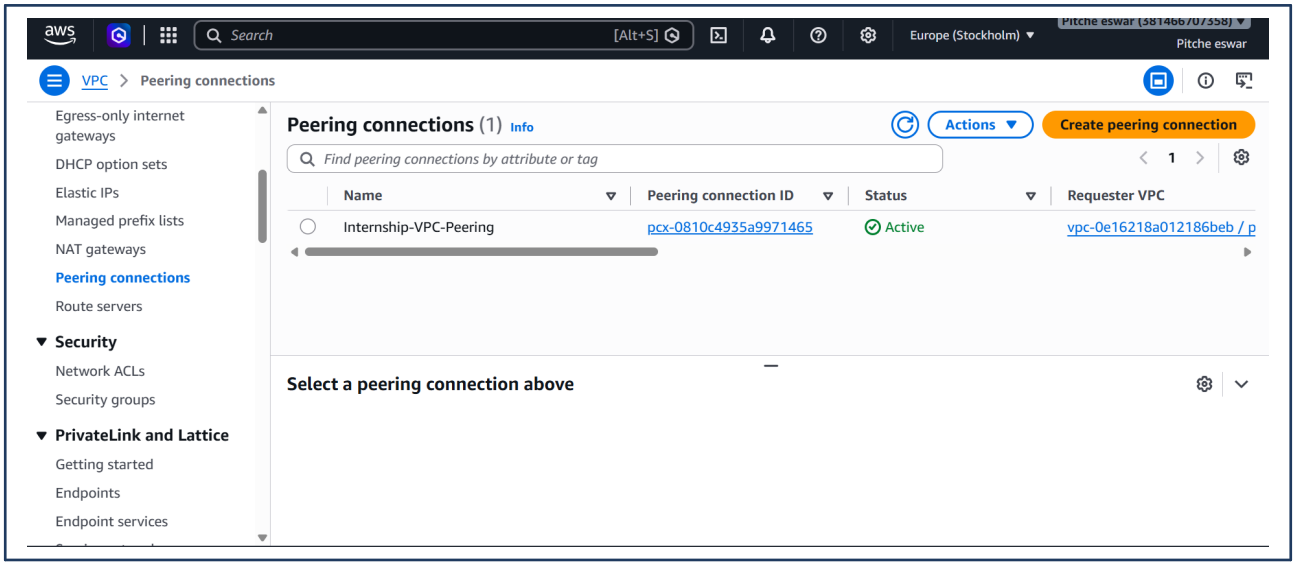
Route tables of Internship-VPC-1 including the public route table (rtb-0c9bce7d5aa32e60c) and the private route table.

Figure 6. Default security group



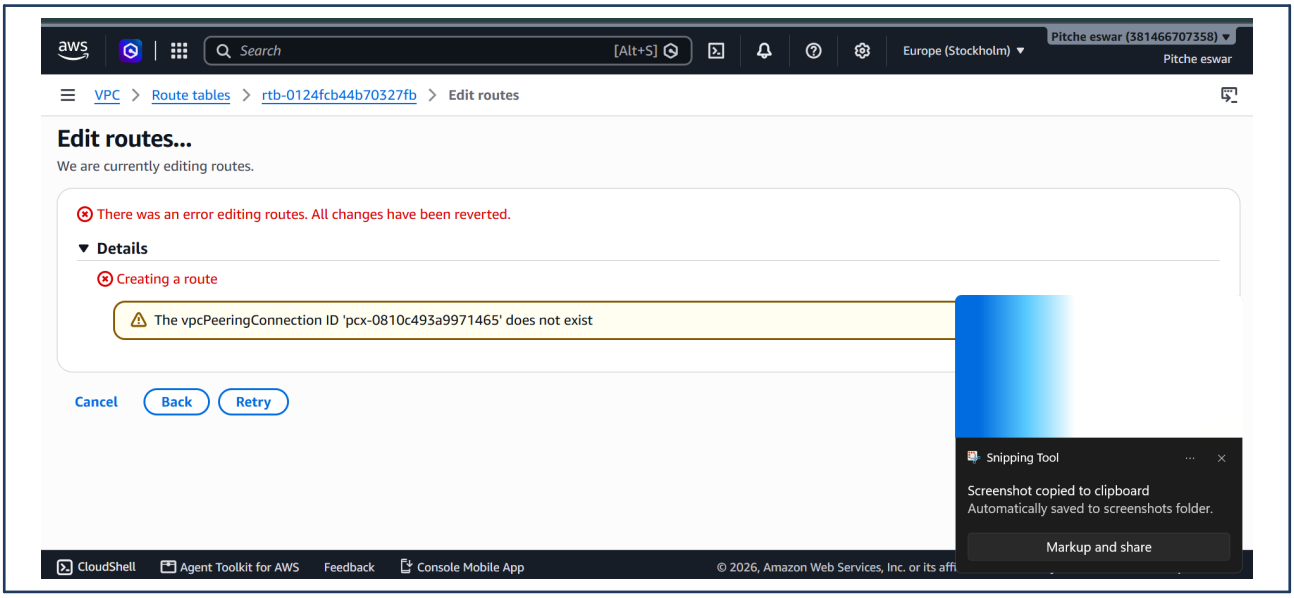
Details of the default security group sg-036490e6042db226f with its inbound rules tab.

Figure 7. Peering connections list



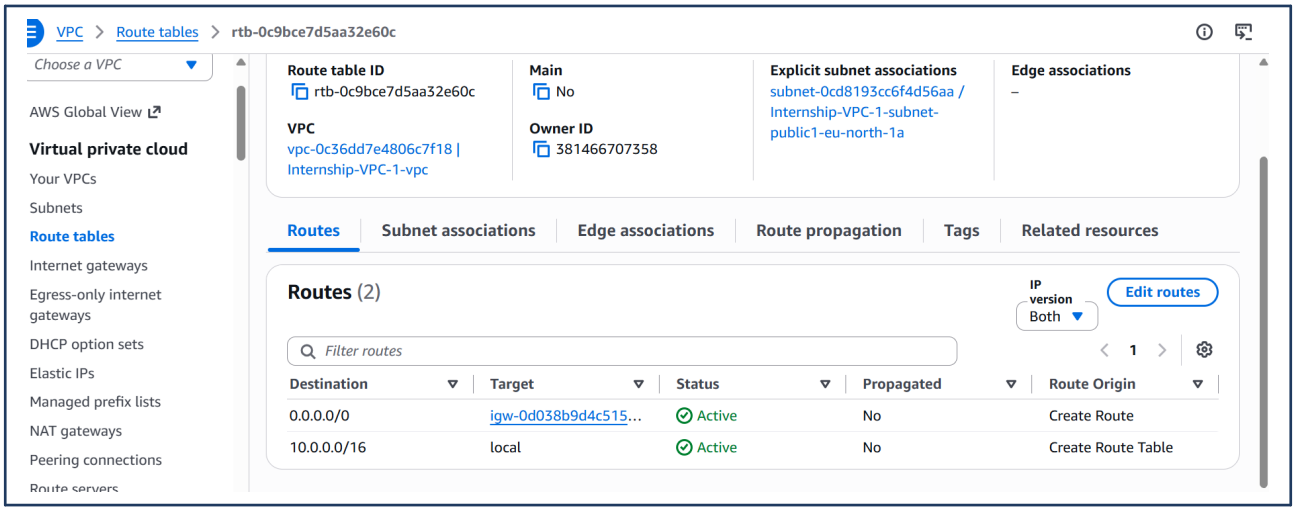
Peering connection pcx-0810c4935a9971465 (Internship-VPC-Peering) listed with status Active.

Figure 8. Error while adding a peering route



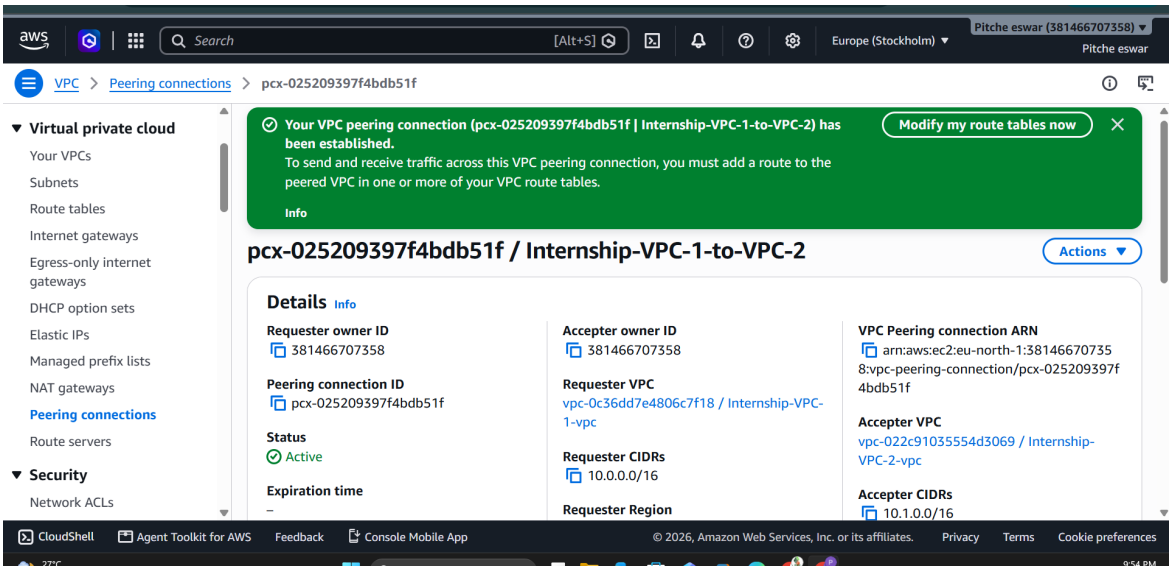
Edit routes failed with the message that the vpcPeeringConnection ID 'pcx-0810c493a9971465' does not exist (route table rtb-0124fcb44b70327fb). The ID was mistyped and the change was reverted.

Figure 9. Public route table before peering route



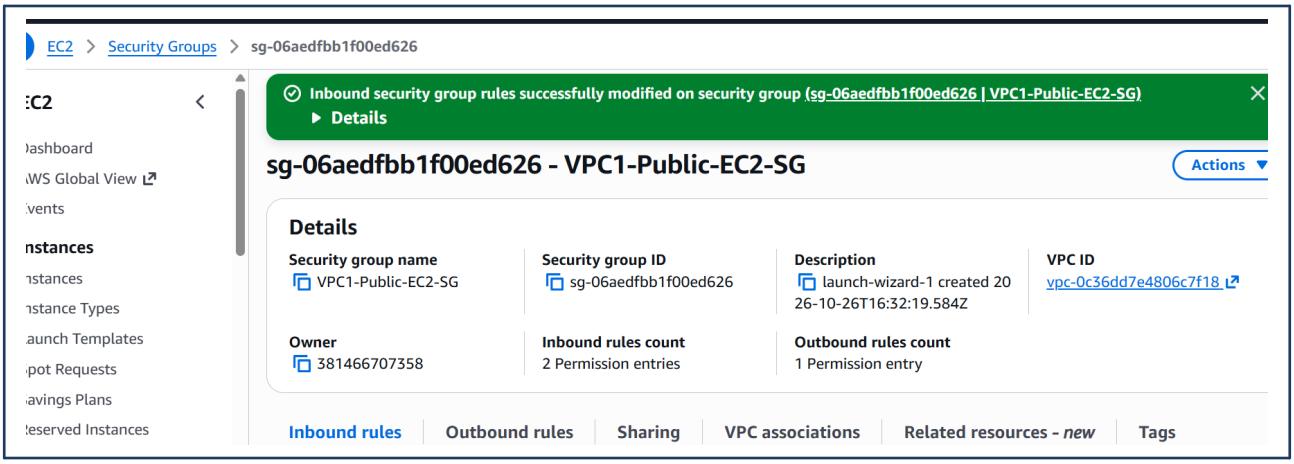
Route table rtb-0c9bce7d5aa32e60c with 2 routes: 0.0.0.0/0 to the internet gateway and 10.0.0.0/16 local.

Figure 10. Peering connection established



pcx-025209397f4bdb51f (Internship-VPC-1-to-VPC-2) is Active between requester CIDR 10.0.0.0/16 and accepter CIDR 10.1.0.0/16.

Figure 11. VPC1-Public-EC2-SG security group



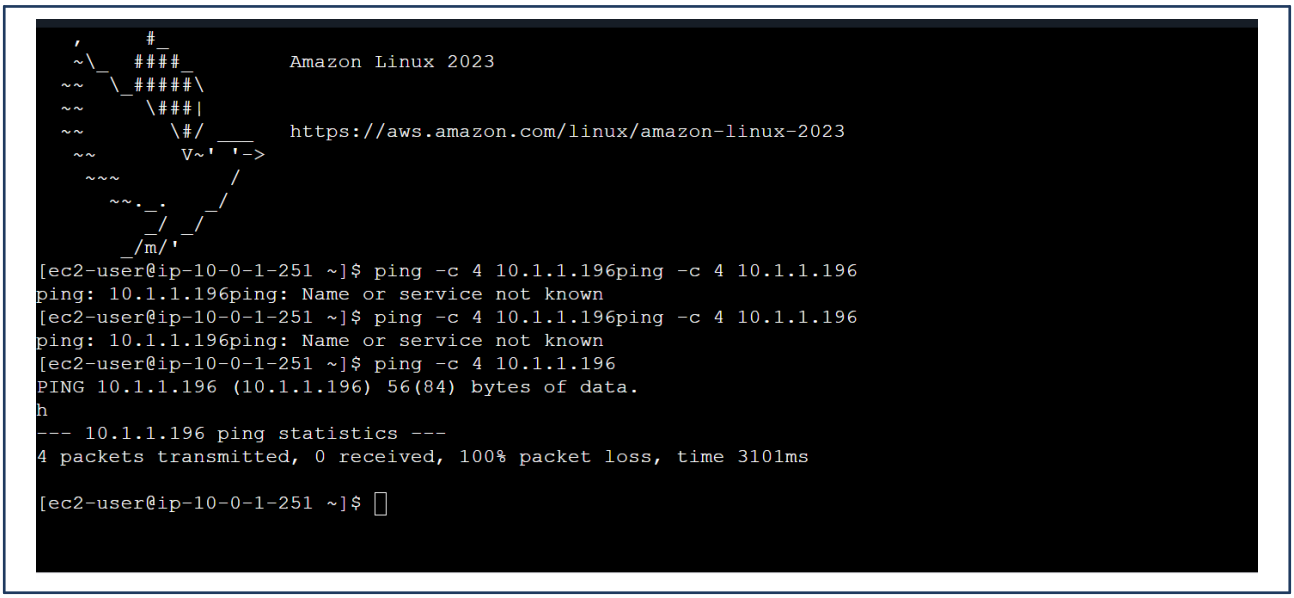
Inbound rules of sg-06aedfbb1f00ed626 successfully modified; the group has 2 inbound rule entries.

Figure 12. First ping attempt from EC2 (typing error)



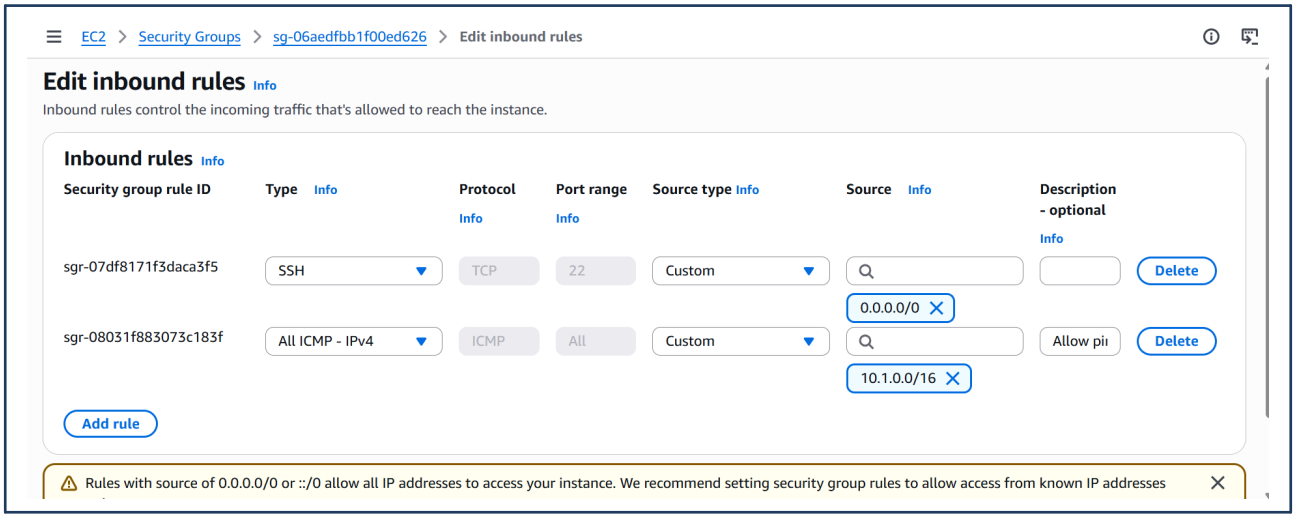
On VPC1-Public-EC2 (private IP 10.0.1.251) the ping command was pasted twice, so it returned 'Name or service not known'.

Figure 13. Ping test failing before the fix



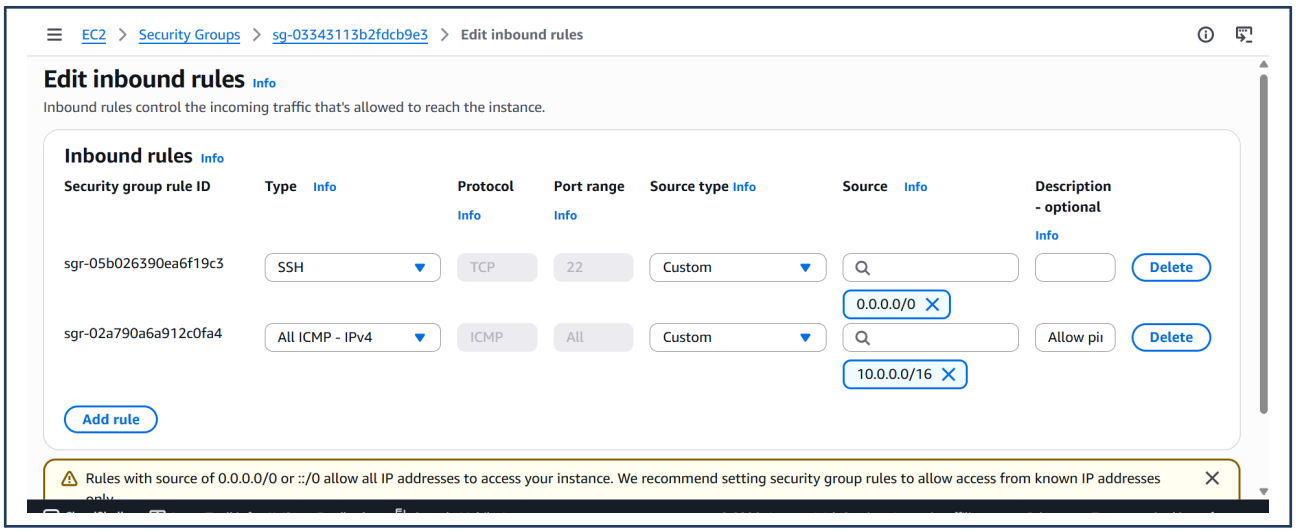
Correct command 'ping -c 4 10.1.1.196' gives 4 packets transmitted, 0 received, 100% packet loss, so the route/security settings were not yet complete.

Figure 14. Inbound rules - VPC 1 instance security group



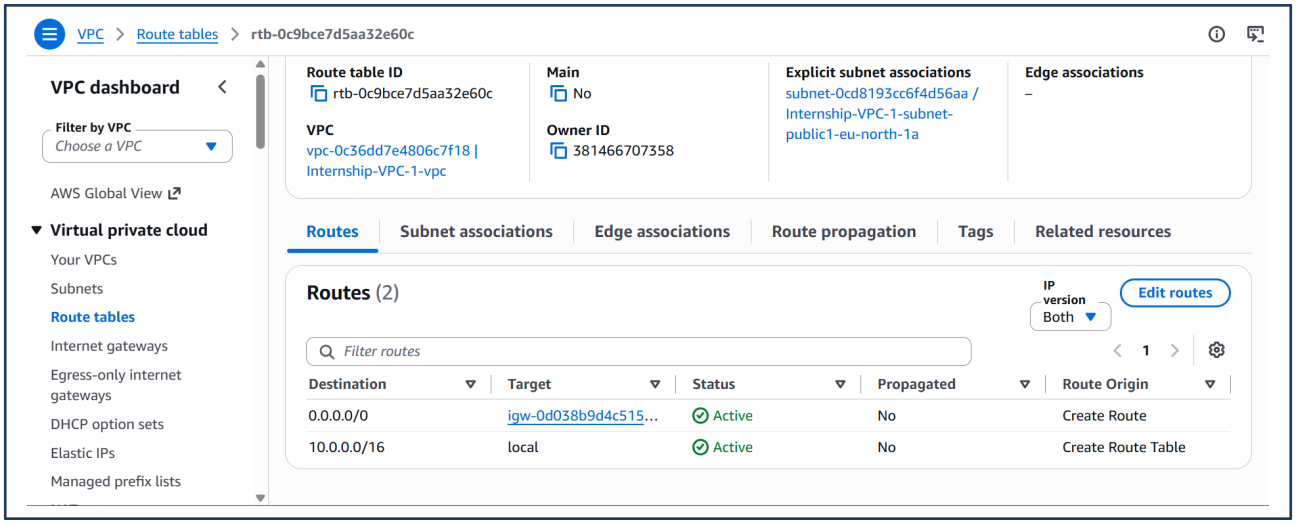
sg-06aedfbb1f00ed626: SSH (TCP 22) from 0.0.0.0/0 and All ICMP-IPv4 from 10.1.0.0/16 (VPC 2 CIDR).

Figure 15. Inbound rules - VPC 2 instance security group



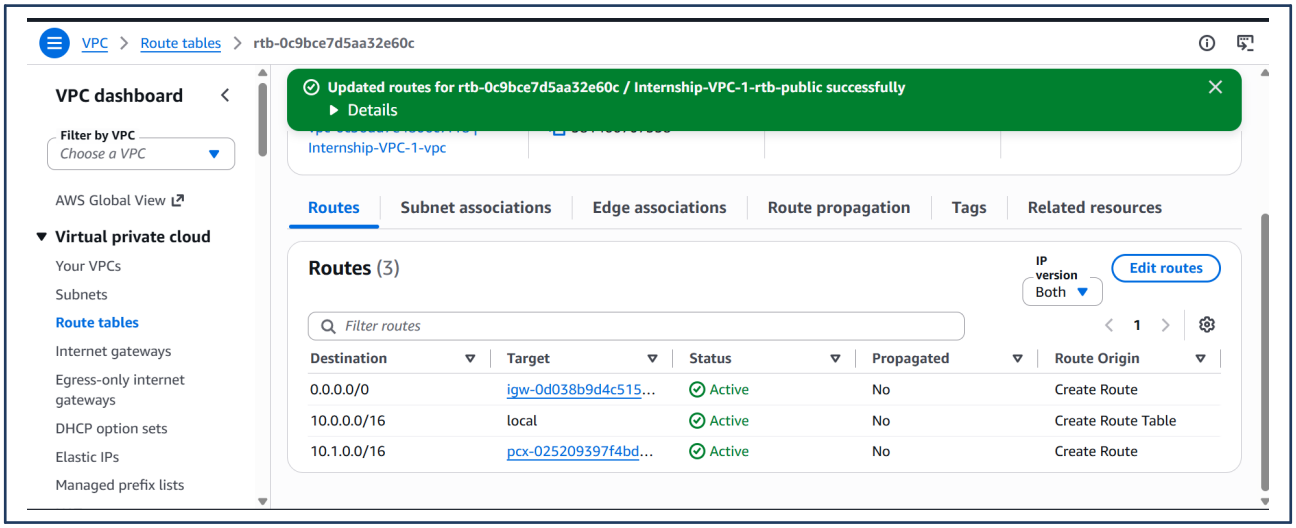
sg-03343113b2fdcb9e3: SSH (TCP 22) and All ICMP-IPv4 from 10.0.0.0/16 (VPC 1 CIDR).

Figure 16. Public route table overview



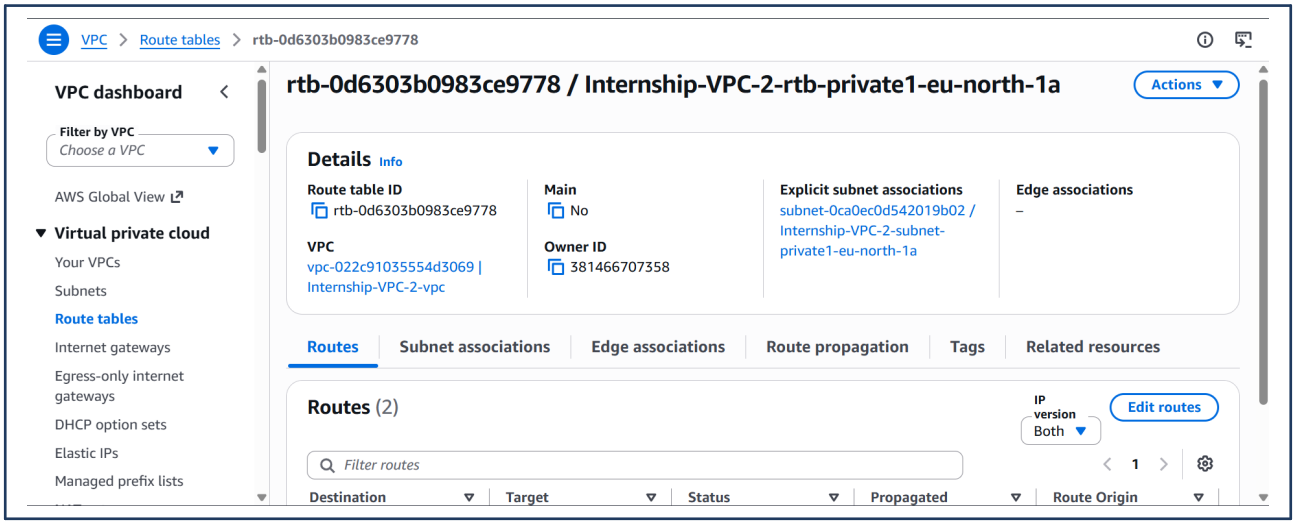
Route table rtb-0c9bce7d5aa32e60c of Internship-VPC-1 before the peering route was added (2 routes).

Figure 17. Peering route added to VPC 1 route table



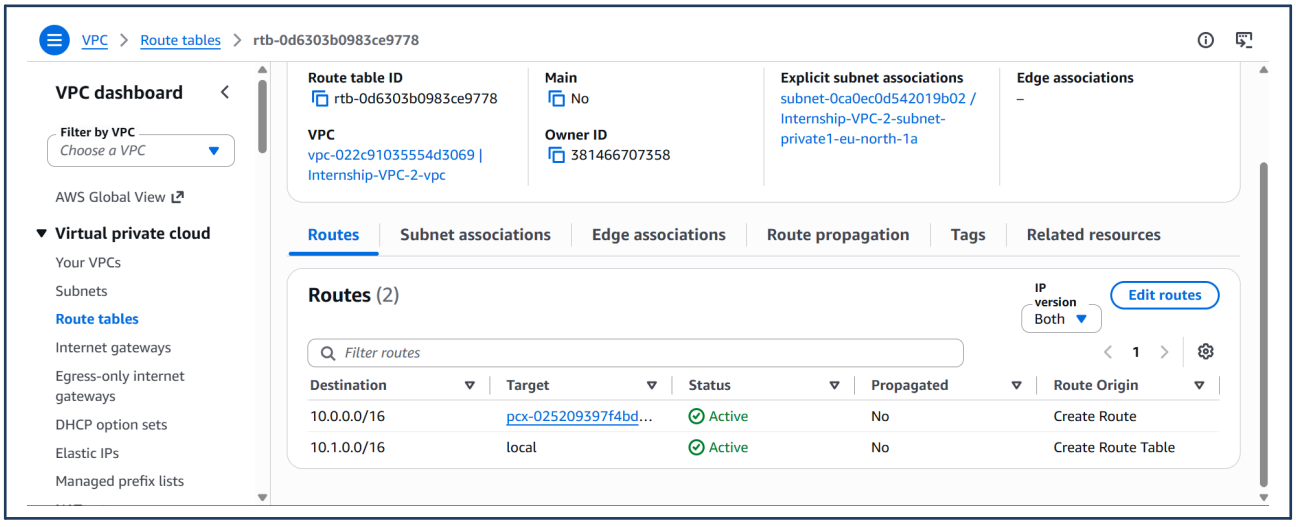
Routes updated successfully: 3 routes, including 10.1.0.0/16 with the peering connection pcx-025209397f4bd... as target.

Figure 18. VPC 2 private route table



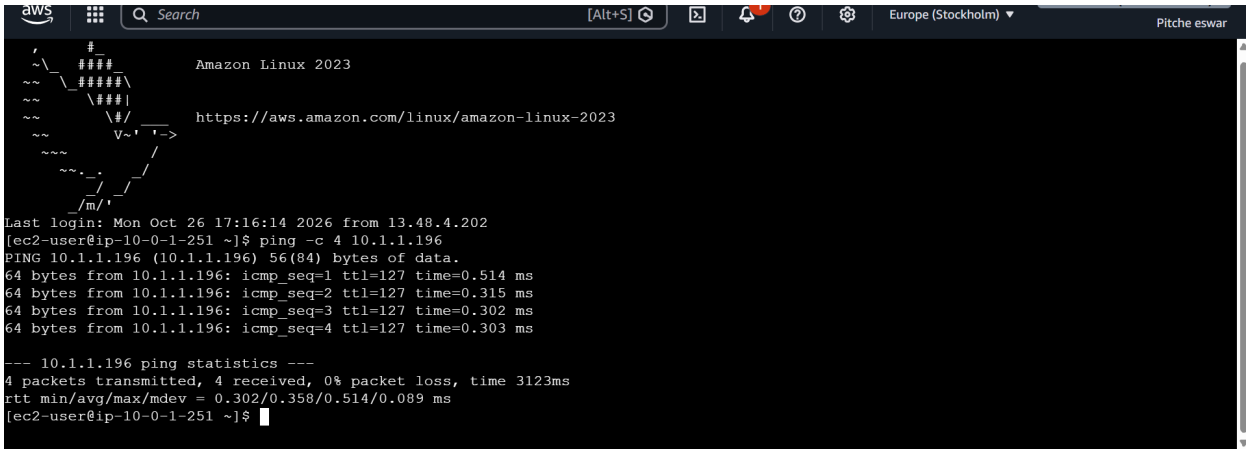
rtb-0d6303b0983ce9778 (Internship-VPC-2-rtb-private1-eu-north-1a) associated with the VPC 2 private subnet.

Figure 19. Peering route added to VPC 2 route table



Routes of rtb-0d6303b0983ce9778: 10.0.0.0/16 to pcx-025209397f4bd... (Active) and 10.1.0.0/16 local.

Figure 20. Successful connectivity test



From 10.0.1.251: ping -c 4 10.1.1.196 gives 4 packets transmitted, 4 received, 0% packet loss, rtt min/avg/max = 0.302/0.358/0.514 ms.